Class 4

	L	M	R
T	1,0	1,2	0,1
B	0,3	0,1	2,0

	M
T	1, 2

The elimination of strictly dominated strategies one by one is called iterative elimination of strictly dominated strategies.

If this process gives you a unique profile, we call it a solution of the game.

Assumption 1' A rational player does not use a weakly dominated strategy

A profile is called rational if it is the unique result of an iterative elimination of weakly dominated strategies.

Second price sealed bid auction

- $N = \{1, 2, \dots, n\}$
- $S_i = [0, \infty)$
- $u_i(b_1, \dots, b_n) = \begin{cases} 0 & \text{if } b_i < \max_{j \in N} b_j \\ \dfrac{v_i - \max_{j \neq i} b_j}{|\{k \mid b_k = \max_{j \in N} b_j\}|} & \text{if } b_i = \max_{j \in N} b_j \end{cases}$

The above game is not a complete information game.

Theorem In the second price sealed bid auction bidding your own valuation weakly dominates all of the strategies.

In other words, for player i,

$$u_i(v_i, b_{-i}) \geq u_i(b_i', b_{-i})$$

$$\forall\, b_{-i} \in [0, \infty)^{n-1}$$

$$\forall\, b_i' \in [0, \infty)$$

Proof See Section 4.6., Theorem 4.15 (Solan's book)

M \ W	F	D
F	(2,1)	(0,0)
D	(0,0)	(1,2)

The max-min approach

1 \ 2	L	R	
T	2,1	2,-20	2
M	3,0	-10,1	-10
B	-100,2	3,3	-100

Suppose player i plays strategy s_i, the the minimum

amount he/she will get is

$$\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$$

$$\underline{v_i} = \max_{s_i \in S_i} \min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i})$$

is called max-min value of player i.

Suppose $s_i^* \in \arg\max_{s_i \in S_i} \left[\min_{s_{-i} \in S_{-i}} u_i(s_i, s_{-i}) \right]$

s_i^* is called a max-min strategy for player i, and by playing a max-min strategy, a player can ensure at least his/her max-min value.

$$u_i(s_i^*, s_{-i}) \geq \underline{v_i} \quad \forall s_{-i} \in S_{-i}$$